

Lesson Objectives

1

Describe the conditions required for the formation of a tropical cyclone.

2

Distinguish between cyclones, hurricanes, typhoons, and tornadoes.

3

Identify cyclone categories and relate them to wind speed.

4

Describe the effects of tropical cyclones on Earth's surface and communities.

CYCLONES

HURRICANES

TYPHOONS

TORNADOES

CATEGORY 1	CATEGORY 2	CATEGORY 3	CATEGORY 4	CATEGORY 5
74-95 mph	96-110 mph	111-129 mph	130-156 mph	157+ mph
INCREASING WIND SPEED				

THE EYE

Calms, clear and warm

SPIRAL RAIN BANDS

Bring heavy rain and strong winds

1

WARM OCEAN WATER (26.5°C OR HIGHER)

2

WARM, MOIST AIR RISES

3

CONDENSATION RELEASES HEAT

4

LOW PRESSURE SYSTEM

5

LOW WIND SHEAR

6

AWAY FROM THE EQUATOR

7

PRE-EXISTING DISTURBANCE

8

CONTINUOUS MOISTURE SUPPLY

1

WARM OCEAN

2

RIISING MOIST AIR

3

LOW PRESSURE

4

CLOUDS AND THUNDERSTORMS

5

SPINNING SYSTEM

6

TROPICAL CYCLONE

WHY DO TROPICAL CYCLONES FORM?

They form when warm oceans, moist air, low pressure, little wind shear, and Earth's rotation work together to build a spinning storm.

FUN FACT!

Tropical cyclones are also called hurricanes (Atlantic & NE Pacific), typhoons (NW Pacific), or cyclones (Indian Ocean & Australia).

The Big Picture

[Click Here](#)

Section A: Watch & Record

As you watch the '[Tropical Cyclones](#)' video, record the following information:

Question	Your Answer
Minimum sea surface temperature needed:	
What happens to warm, moist air as it rises?	
Direction of rotation in the Southern Hemisphere:	
The name of the calm centre of the storm:	
Wind speed to be classified as a cyclone (km/h):	
What happens when a cyclone moves over land?	

How a Tropical Cyclone Forms

Conditions needed for formation

1

WARM OCEAN WATER (26.5°C OR HIGHER)

2

WARM, MOIST AIR RISES

3

CONDENSATION RELEASES HEAT

4

LOW PRESSURE SYSTEM

5

LOW WIND SHEAR

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AWAY FROM THE EQUATOR

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PRE-EXISTING DISTURBANCE

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CONTINUOUS MOISTURE SUPPLY

THE JOURNEY: FROM BEGINNING TO TROPICAL CYCLONE

1 WARM OCEAN 2 RIISING MOIST AIR 3 LOW PRESSURE 4 CLOUDS AND THUNDERSTORMS 5 SPINNING SYSTEM 6 TROPICAL CYCLONE

WHY DO TROPICAL CYCLONES FORM?

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Section B: Terminology Comparison

Complete the table to distinguish between different types of spinning storms:

Term	Where Used / Formed	Key Characteristic
Cyclone		
Hurricane		

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Typhoon
Tornado



Section C: Cyclone Categories

BTN: Tropical Cyclones

[Watch this](#)

As discussed in class, complete the table relating cyclone categories to wind speeds:

Category	Strongest Gusts (km/h)	Potential Impact
1		
2		
3		
4		
5		



Section D: Case Study Comprehension

[Audio Source](#)

[Video Source 1](#) (Alfred)

[Video Source 2](#) (Debbie)

After researching your case study (Yasi or Larry), answer the following:

Yasi

Larry

[Website Source](#)

[Website Source](#)

1. What was the maximum wind speed recorded for this cyclone? What category was it?

2. Identify one major effect this cyclone had on the Earth's surface (environment).

3. Describe the effects of this cyclone on the local community (buildings, people).

4. How did the community prepare for this event?

5. Why was the number of fatalities so low?